

Lukas Lazarek

Boston, MA

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EDUCATION

Northwestern University, Evanston, IL

Ph.D. in Programming Languages

Jun 2024

University of Massachusetts Lowell, Lowell, MA

B.S. in Computer Science

Minor in Robotics

May 2018

RESEARCH AND WORK EXPERIENCE

Brown University, Providence, RI

Postdoctoral Research Associate

Aug 2024–Present

Center for Neighborhood Technology, Chicago, IL

Urban Analytics Intern

Mar 2023–Jun 2023

Carnegie Mellon University ISR, Pittsburgh, PA

Undergraduate Researcher

May 2017–Apr 2018

Symbotic, Wilmington, MA

Software Engineer Intern

Jan 2017–May 2017

PUBLICATIONS

Conference Publications

Lukas Lazarek, Evangelos Lamprou, George Kapetanakis, Anirudh Narsipur, Eric Zhao, Zhiwen Zheng, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis. 2026. “Ahead-of-time Analysis of Shell Program Effects.” *ACM Symposium on Operating Systems Principles (SOSP)*. **Awarded best paper.**

Zekai Li, **Lukas Lazarek**, Evangelos Lamprou, George Kapetanakis, Konstantinos Mamouras, Nikos Vasilakis. 2026. “RT: Regular Types for the Streaming Shell.” *USENIX Symposium on Operating Systems Design and Implementation (OSDI)*.

Evangelos Lamprou, Ethan Williams, Georgios Kaoukis, Zhuoxuan Zhang, Michael Greenberg, Konstantinos Kallas, **Lukas Lazarek**, Nikos Vasilakis. 2025. “The Koala Benchmarks for the Shell: Characterization and Implications.” *USENIX Annual Technical Conference (ATC)*. **Awarded best paper.**

Lukas Lazarek, Seong-Heon Jung, Evangelos Lamprou, Zekai Li, Anirudh Narsipur, Eric Zhao, Michael Greenberg, Konstantinos Kallas, Konstantinos Mamouras, Nikos Vasilakis. 2025. “From Ahead-of- to Just-in-Time and Back Again: Static Analysis for Unix Shell Programs.” *ACM SIGOPS Workshop on Hot Topics in Operating Systems (HotOS)*.

Lukas Lazarek, Ben Greenman, Matthias Felleisen, Christos Dimoulas. 2023. “How to Evaluate Blame for Gradual Types, Part 2.” *ACM SIGPLAN International Conference on Functional Programming (ICFP)*. Series 7.

Lukas Lazarek, Ben Greenman, Matthias Felleisen, Christos Dimoulas. 2021. “How to Evaluate Blame for Gradual Types.” *ACM SIGPLAN International Conference on Functional Programming (ICFP)*. Series 5.

Lukas Lazarek, Alexis King, Samanvitha Sundar, Robert Bruce Findler, Christos Dimoulas. 2020. “Does Blame Shifting Work?.” *ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*. Series 4.

Chu-Pan Wong, Jens Meinicke, **Lukas Lazarek**, Christian Kästner. 2018. “Faster Variational Execution with Transparent Bytecode Transformation.” *ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*. Series 2.

Journal Publications

Evangelos Lamprou, Ethan Williams, Georgios Kaoukis, Zhuoxuan Zhang, Michael Greenberg, Konstantinos Kallas, **Lukas Lazarek**, Nikos Vasilakis. 2026. “The Koala Benchmarks for the Shell: Characterization, Implications, and Extensions.” *ACM Transactions on Computer Systems (TOCS)*. Vol. TBD, No. TBD.

Ben Greenman, **Lukas Lazarek**, Christos Dimoulas, Matthias Felleisen. 2022. “A Transient Semantics for Typed Racket.” *Programming*. Vol. 2, No. 6.

TEACHING EXPERIENCE

Brown University

Mentor: ATLAS graduate and undergraduate students Aug 2024-Present
Mentor over a dozen students (PhD, masters, and undergraduate) across all phases of research: project exploration, scoping, implementation, evaluation, writing, submission, and dissemination.

Brown Systems Week instructor, Analysis of Shell Programs Sum25, Sum26
Taught annual week-long mini-course in the Brown Systems Week

Brown Systems Transforming Systems Research Seminar F24, F25
Co-designed and taught modules on topics including rigorous experiment design, presenting data effectively, and formal semantics

Northwestern University

Instructor, Programming Languages (CS321) Summer'23
Taught the undergraduate programming languages course

Teaching Assistant (TA), Programming Languages (CS321) S23, W23, F23, S24
Co-developed, taught workshop series on pragmatics of language design

TA, Proving Properties of Programs with Mechanized Logic (CS393) Fall'21
Developed, taught course module about weakest precondition logics

TA, Software Construction (CS393)	F19, S21, S22
<i>Developed integral CI-based autograder feedback infrastructure</i>	
TA, Intensive Program Design (CS495)	Fall'18
Mentor: Northwestern PL students	2023-2024
<i>Co-mentored one undergraduate student in research, who went on to start a PhD</i>	

FUNDING

NSF grant #2525351: FMitF track I: Reasoning About Shell Scripts and Their Effects in Context	2026
<i>Co-PI with Nikos Vasilakis, Konstantinos Kallas, and Michael Greenberg</i>	

INVITED TALKS

Semantics-Driven Static Analysis for UNIX Shell programs	Feb 2026
<i>Northeastern University</i>	
Mutate: Inject Bugs into Your Programs!	Oct 2023
<i>Thirteenth RacketCon</i>	

AWARDS

SOSP Best Paper Award - Ahead-of-time Analysis of Shell Program Effects	2026
ATC Best Paper Award - The Koala Benchmarks for the Shell	2025
Northwestern Computer Science PhD student best research award	2021
NSF GRFP honorable mention	2020
ACM SRC 1st place award (undergraduate category, at SPLASH'17)	2017

PROFESSIONAL ACTIVITIES

PLDI Program Committee	2026
ICFP Program Committee	2026
JFP reviewer	2020
ICFP Artifact Evaluation Committee	2020
POPL student volunteer	2020
ICFP student volunteer	2018

DEPARTMENT/GROUP SERVICE

Brown University

Co-leader of ATLAS media communication team	May 2026–Present
Organizer of ATLAS social events	Jun 2025–Present

Northwestern University

Computer Science Social Initiative board	Apr 2020–Jun 2024
Computer Science PhD Advisory Council board	Oct 2019–Dec 2020

OUTREACH ACTIVITIES

The Bloc Chicago volunteer, co-leader of tech program <i>Creating a space for youth to have fun and develop a creative relationship with technology</i>	Jun 2021–Jun 2024
EvanSTEM mentor	Oct 2020–Apr 2021
Boys and Girls Club of Greater Lowell mentor	Oct 2016–Jun 2018